

Parallel Processing Concepts

Q. What is Parallel Processing Discuss the need for parallel processing, its types, and its primary advantages in computer architecture.

[DEF] Definition to Remember

Parallel Processing is an architectural technique where a computer executes multiple instructions or processes multiple pieces of data simultaneously using multiple processing units (like multiple ALUs or cores). It shifts computing from sequential execution to concurrent multitasking.

1. Need for Parallel Processing

Historically, processors became faster by increasing the clock frequency. However, this approach hit physical limits (the power wall), causing extreme heat and power consumption.

2. Types of Parallelism

1. Instruction-Level Parallelism (ILP):

3. Advantages of Parallel Processing

- * Increased Speed / Throughput: Executing tasks simultaneously drastically reduces total execution time.

[MUST-WRITE] Must-Write Points (for 10 marks)

1. Parallel Processing: Executing multiple instructions/data simultaneously to increase speed.
2. Need: Overcomes physical clock speed limits (heat/power wall) and meets high-computation demands (AI, graphics).
3. Instruction-Level Parallelism (ILP): Uses pipelining and superscalar techniques.
4. Task-Level Parallelism: Runs different programs concurrently using multi-core CPUs.
5. Data-Level Parallelism: Applies one operation to vast amounts of data concurrently (GPUs).
6. Advantages: Massive speed/throughput increase, better resource utilization.
7. Advantages: Energy-efficient (compared to extreme overclocking) and provides fault tolerance.

[RECALL] Quick Recall

Parallel Processing (Concurrent execution) -> Need (Hit physical clock limits, AI/Graphics demand) -> Types: ILP (Pipelining), Task-Level (Multi-core), Data-Level (GPU) -> Pros: Fast, Energy Efficient, Fault Tolerant